Università degli studi di Padova

Stochastic Methods for Finance

Report 6: Monte Carlo simulations for Pricing Options

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1 Introduction

This report investigates the application of Monte Carlo (MC) simulation for pricing financial derivatives within the Black-Scholes framework. We assume an underlying asset price process $\{S_t\}_{t\geq 0}$ governed by the Geometric Brownian Motion (GBM) stochastic differential equation:

$$dS_t = rS_t dt + \sigma S_t dW_t$$

Under the assumptions of a frictionless market, constant risk-free rate r, and constant volatility σ, we evaluate the numerical precision of different discretization schemes. After benchmarking these methods against analytical formulas, the framework is extended to path-dependent Asian and Lookback options. Finally, the study addresses multi-asset complexity by pricing a "Worst-Of" Certificate on a basket of three equities.

2 Asset Path Simulation and Visualization

The initial phase of the study involves simulating $N = 100$ independent trajectories for the underlying asset over a one-year horizon ($T = 1$), discretized into $N_{steps} = 252$ daily intervals. Based on the parameters $S_0 = 100$, $r = 2\%$, and $\sigma = 40\%$, the paths were generated using two primary numerical approaches.

Euler Discretization The Euler method is a first-order numerical approximation that linearizes the SDE over the interval Δt. It is expressed as:

$$S_{t+\Delta t} = S_t + S_t(r\Delta t + \sigma\sqrt{\Delta t}Z)$$

Exact Solution (Closed-Form Concatenation) By applying Ito's Lemma to the log-price, we obtain the exact solution. This method preserves the log-normal distribution of the asset and is expressed as:

$$S_{t+\Delta t} = S_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}Z\right)$$

In both equations, $Z \sim N(0,1)$ represents a standard normal random variable generated via the inverse transform method.

The visualizations present the characteristic "fan" shape of Geometric Brownian Motion, where the dispersion of possible asset prices increases over time. Visually, both schemes produce similar diffusion patterns. However, the Exact Solution is theoretically superior as the exponential term ensures that $S_t > 0$ for all t. Despite the stochastic noise, the paths center around a slight upward trend, driven by the 2% risk-free drift.

3 Vanilla Option Pricing

This section evaluates the performance of three Monte Carlo pricing methods for European Call and Put options. The objective is to analyze how different discretization schemes and

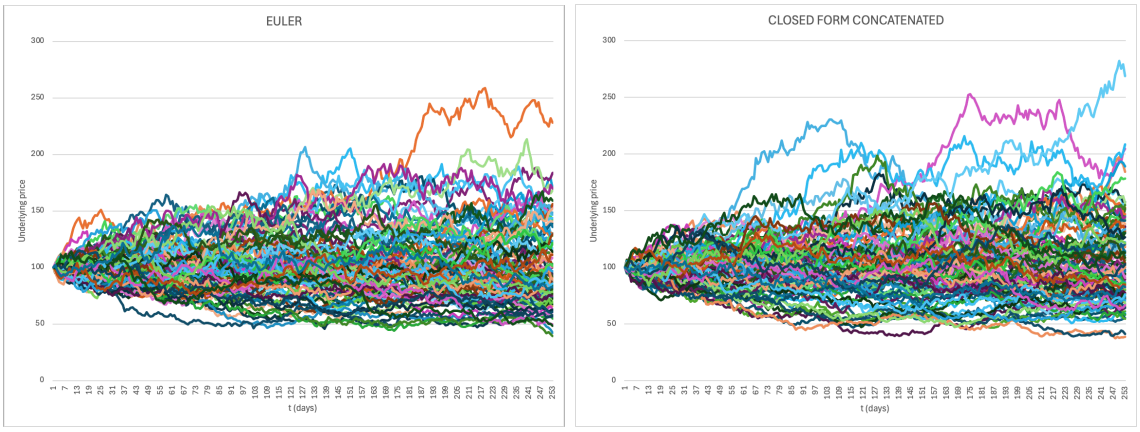

Figure 1: Simulated trajectories. $S_0 = 100$, $r = 0.02$, $\sigma = 0.4$, $T = 1$, $N_{steps} = 252$, $N_{paths} = 100$, $K = 110$.

the number of simulation paths (N_{paths}) influence the accuracy and stability of the price estimate.

The options were priced using the following three numerical approaches

- One-Step Closed Form: This method simulates only the terminal price S_T using the exact solution of the GBM SDE. Since it bypasses the intermediate path, it is computationally efficient and free from discretization error.
- Multi-Step Euler: This scheme approximates the path over 252 daily steps using the linear approximation.
- Multi-Step Exact (Concatenated): This method simulates the full path using the exact solution at every daily interval.

To evaluate the quality of the results, the Monte Carlo prices were compared against the Black-Scholes (B&S) Formula benchmark. Two primary metrics were used: absolute error (the difference between the MC price and the B&S benchmark) and standard mean error (a measure of the statistical "noise" in the simulation, calculated as $\frac{\sigma_{payoffs}}{\sqrt{N_{paths}}}$).

As N_{paths} increases from 100 to 10,000, simulated Call and Put prices clearly tend to stabilize and converge toward the horizontal B&S benchmark line.

The "Absolute Error" bar charts demonstrate a sharp decline in error as the sample size grows. This follows the theoretical $O(1/\sqrt{N})$ convergence rate characteristic of Monte Carlo methods. The Standard Error (represented in error bars) decreases consistently with the square root of the number of paths.

4 Path-Dependent Options: Asian and Lookback

This section explores the pricing of exotic derivatives where the payoff is determined by the asset's behavior throughout the life of the option. Unlike vanilla options, these require a multi-step simulation to capture the intermediate price history.

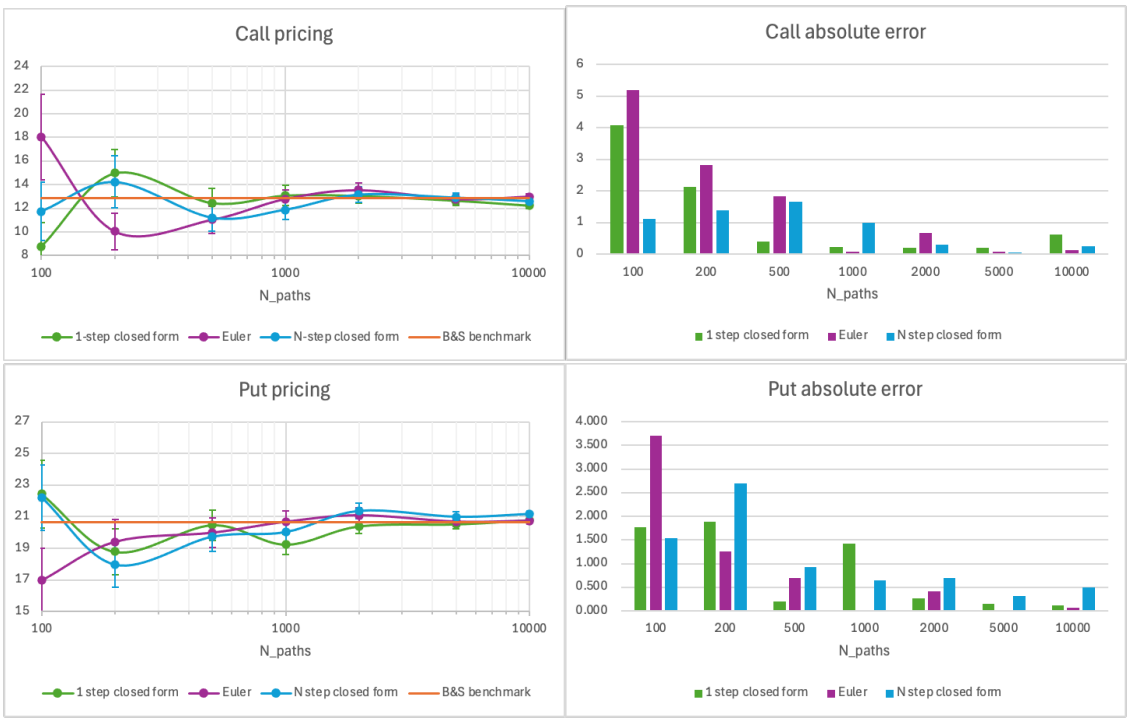

Figure 2: Vanilla Option prices. Absolute error from Black-Scholes benchmark. $S_0 = 100$, $r = 0.02$, $\sigma = 0.4$, $T = 1$, $N_{steps} = 252$, $K = 110$.

4.1 Asian Option

Asian options mitigate the risk of price manipulation or extreme volatility near maturity by basing the payoff on the average price of the underlying asset.

The payoff equations for continuous averaging are:

- Asian Call: $\max\left(\frac{1}{T}\int_0^T S_t dt - K, 0\right)$
- Asian Put: $\max\left(K - \frac{1}{T}\int_0^T S_t dt, 0\right)$

In our discrete model, the integral is approximated by the arithmetic mean of the $N = 252$ simulated price points: $\bar{S} = \frac{1}{N}\sum_{i=1}^{N} S_{t_i}$

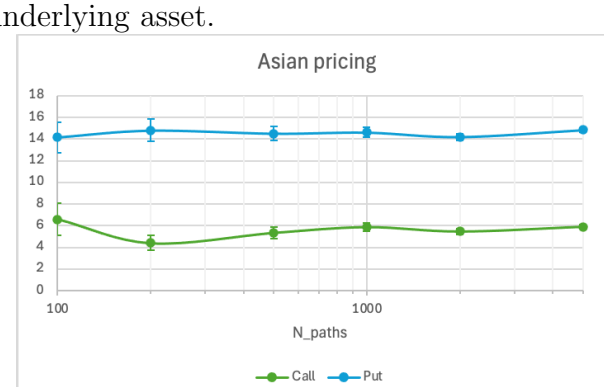

Figure 3: Asian Option prices. $S_0 = 100$, $r = 0.02$, $\sigma = 0.4$, $T = 1$, $N_{steps} = 252$, $K = 110$.

Asian options are significantly cheaper than their vanilla counterparts (Call ≈ 5.89 vs. Vanilla 12.85). This is because the averaging process reduces the effective volatility of the payoff, making extreme out-of-the-money outcomes less likely.

4.2 Lookback Option

Lookback options allow the holder to "look back" over time and exercise the option based on the maximum or minimum price achieved during the period. The payoff equations for continuous averaging are:

- Lookback Call: $S_T - \min(S_t, 0 \leq t \leq T)$
- Lookback Put: $\max(S_t, 0 \leq t \leq T) - S_T$

Lookback options are the most expensive derivatives in this study (Call $\approx$ 27.35, Put $\approx$ 32.42).

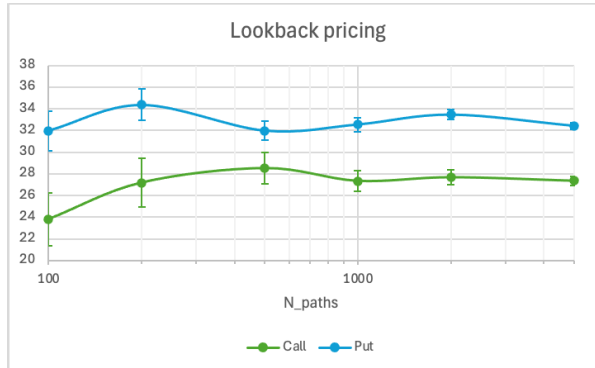

Figure 4: Lookback Option prices. $S_0 = 100$, $r = 0.02$, $\sigma = 0.4$, $T = 1$, $N_{steps} = 252$, $K = 110$.

Their payoff structure guarantees the best possible price relative to the minimum or maximum, providing much higher value than standard vanillas.

The Lookback charts show slightly higher variance in the early stages ($N < 500$) compared to Asian options. This is expected, as the payoff depends on a single extreme point (Max or Min) rather than a smoothed average, making the simulation more sensitive to outliers in smaller sample sizes.

5 Multi-Asset "Worst-Of" Certificates

The final stage of the study involves pricing structured products written on a basket of three European equities: L'Oréal (OR.PA), a global leader in consumer goods and luxury; Air Liquide (AI.PA), a major industrial gases firm; and BNP Paribas (BNP.PA), a leading financial institution. This selection provides a diversified exposure across the consumer, industrial, and banking sectors.

To price these products, a historical covariance matrix was estimated using one year of daily returns. Because the assets are not independent, a Cholesky Decomposition (L) was applied to the annualized covariance matrix $\Sigma = LL^T$. This allowed for the simulation of correlated paths by transforming independent standard normal variables Z into correlated shocks $\epsilon = LZ$.

Two types of "Worst-Of" structures were evaluated, both defined by the performance of the weakest asset in the basket: $R_{min} = \min(S_{1,T}/S_{1,0}, S_{2,T}/S_{2,0}, S_{3,T}/S_{3,0})$.

- Worst-Of Tracker: A linear product where the payoff is strictly the performance of the worst asset multiplied by the notional:

$$Payoff = \text{Notional} \times R_{min}$$

- Worst-Of with Barrier: A capital-protected product (European barrier) that returns

the full notional unless the worst asset falls below a specified threshold ("barrier"):

$$Payoff = \begin{cases} \text{Notional} & \text{if } R_{min} \geq \text{Barrier} \\ \text{Notional} \times R_{min} & \text{if } R_{min} < \text{Barrier} \end{cases}$$

To ensure the simulation reflects current market dynamics, historical data and risk-free benchmarks were retrieved from Refinitiv. Initial spot prices (S_0) were set based on the most recent closing prices at the time of retrieval. The 12-month Euribor rate ($\approx 2.655\%$) was used as the proxy for the risk-free rate, consistent with the one-year investment horizon ($T = 1$). A one-year window of historical daily returns was analyzed to compute the daily covariance matrix. To align with the annual parameters of the Black-Scholes model, the daily variances and covariances were scaled by a factor of 252 (the number of trading days in a year) to produce the Annual Volatility Matrix Σ:

$$\Sigma_{annual} = \Sigma_{daily} \times 252$$

The simulation was executed with $N_{steps} = 252$ (daily steps) to accurately capture potential barrier breaches, a Notional (Par Value) of 100, and a Downside Barrier set at 60%.

The Tracker price is significantly lower than the individual stock expectations because the probability of at least one asset underperforming is higher than that of a single asset. The Barrier version is more expensive because it provides "conditional protection": the investor only loses principal if the worst asset drops by more than 40%.

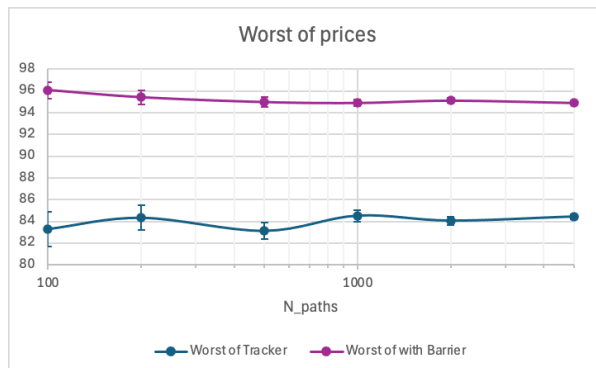

Figure 5: Worst-Of Certificate prices.

In the financial industry, these products (often called "Barrier Reverse Convertibles") are typically sold at a price equal to the Notional (in our case 100). However, as our simulation shows, the "fair value" of the repayment part is only 94.92. This creates a surplus for the bank: the issuer uses the remaining 5.08 to fund periodic coupon payments and cover management fees. If all three stocks remain above the 60% barrier at maturity, the investor receives their full Par Value (100) back, plus all the coupons collected during the year. If any stock finishes below the barrier, the investor receives the "Worst-Of" performance, potentially suffering a significant loss of principal.